

Package ‘SyncERdata’

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Title Example Datasets for 'SyncER'

Version 1.0.0

Description Bundled example datasets used by the 'SyncER' package vignette and test suite: synthetic radiocarbon-dated event records for five cores and the corresponding completed 'rbacon'/'rplum' age-depth model output (raw, synchronized, and synchronized-without-14C variants). These data let 'SyncER' demonstrate and test its full workflow reproducibly, without requiring users to install 'rbacon'/'rplum' or re-run Bayesian age-depth modelling. This package contains the synthetic data from Wils & Ramisch (2026) <[doi:10.1038/s41598-026-67943-7](https://doi.org/10.1038/s41598-026-67943-7)>.

License GPL (>= 3)

URL <https://github.com/katleenwils/SyncERdata>

BugReports <https://github.com/katleenwils/SyncERdata/issues>

Depends R (>= 3.5.0)

Encoding UTF-8

LazyData true

LazyDataCompression xz

Config/roxygen2/version 8.0.0

NeedsCompilation no

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bacon_out_files	<i>Bacon/Plum age-depth model output for the example records</i>
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Description

Completed **rbacon/rplum** age-depth model output (.out files) for the five example records (see [record_data_input](#)), in three variants: raw, synchronized, and synchronized-without-14C.

Usage

```
bacon_out_files
```

Format

A named list with one element per relative output-file path (e.g. "core1/core1_121.out", "core1_synced/core1_synced_121.out"). Each element is a character vector holding the lines of the corresponding .out file, exactly as written by **rbacon/rplum**.

Details

Bayesian age-depth modelling is not deterministic across runs, so bundling the completed output lets **SyncER**'s vignette and tests demonstrate and exercise the full workflow – including the synchronization and re-modelling steps – quickly and reproducibly, without requiring **rbacon/rplum** to be installed or re-run.

To restore the files to disk (as **SyncER** expects them), write each element to a file at its relative path within some target directory, e.g.:

```
for (rel_path in names(bacon_out_files)) {
  full_path <- file.path(target_dir, rel_path)
  dir.create(dirname(full_path), recursive = TRUE, showWarnings = FALSE)
  writeLines(bacon_out_files[[rel_path]], full_path)
}
```

Source

Generated for the SyncER manuscript (Wils & Ramisch 2026) using **rbacon/rplum** on the synthetic record_data_input example data.

Examples

```
data(bacon_out_files)
names(bacon_out_files)[1]
head(bacon_out_files[[1]])
```

out_data_ages	<i>Posterior age samples for the example records (raw)</i>
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Description

Posterior age samples per event horizon, extracted from [bacon_out_files](#) for the raw (non-synchronized) age-depth models of the five example records.

Usage

```
out_data_ages
```

Format

A named list of five data frames (core1-core5), one per record, each holding posterior age samples with one column per event horizon.

See Also

[out_data_ages_synced](#), [bacon_out_files](#)

Examples

```
data(out_data_ages)
names(out_data_ages)
head(out_data_ages$core1)
```

out_data_ages_synced	<i>Posterior age samples for the example records (synchronized)</i>
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Description

Posterior age samples per event horizon, extracted from [bacon_out_files](#) for the synchronized age-depth models of the five example records (i.e. after applying `SyncER::synchronize_ages()`).

Usage

```
out_data_ages_synced
```

Format

A named list of five data frames (core1_synced-core5_synced), one per record, each holding posterior age samples with one column per event horizon.

See Also

[out_data_ages](#), [bacon_out_files](#)

Examples

```
data(out_data_ages_synced)
names(out_data_ages_synced)
head(out_data_ages_synced$core1_synced)
```

record_data_input	<i>Synthetic record input data (raw CSVs)</i>
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Description

Five synthetic radiocarbon-dated event records (core1-core5), one CSV file per record, shipped as raw files rather than as an R object so that `SyncER::read_record_data()` can read them exactly as it would a user's own `record_data_input` folder.

Details

Access the folder with:

```
system.file("extdata", "record_data_input", package = "SyncERdata")
```

Each file contains, per dated sample and event horizon in the record: depth, age, error, calibration curve indicator, and event label.

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